

A4.3.6 Reinforcement Learning



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A4.3.6 covers reinforcement learning, where agents learn decisions by interacting with their environment, based on cumulative reward, actions, states, rewards, policies, and the exploration vs. exploitation trade-off. A4.3.7 describes genetic algorithms, applied in optimisation problems like route planning (travelling salesperson problem), using concepts like population, fitness, selection, crossover, mutation, evaluation, and termination

Introduction

We have explored how machines can learn from labeled data (Supervised) and unlabeled data (Unsupervised). Now we venture into a third, powerful paradigm: **Reinforcement Learning (RL)**. This is how we teach a machine to learn from scratch by interacting with an environment, much like how a baby learns to walk or how you learn to play a new video game.

The Agent-Environment Interaction

The core of Reinforcement Learning is the **agent-environment interaction loop**. Instead of a static dataset, we have an active **agent** that learns by "doing."

- **Analogy:** Think of training a dog. The dog is the **agent**. The room you are in is the **environment**.

The process is a continuous cycle:

1. The agent observes the current **state** of the environment.
2. Based on its strategy (its **policy**), the agent chooses an **action** to take.
3. The environment reacts to the action. It transitions to a new state and gives the agent a **reward** (or penalty).
4. The agent receives this new state and reward, learns from the feedback, and the cycle repeats.

Reinforcement Learning (RL): The Agent-Environment Loop

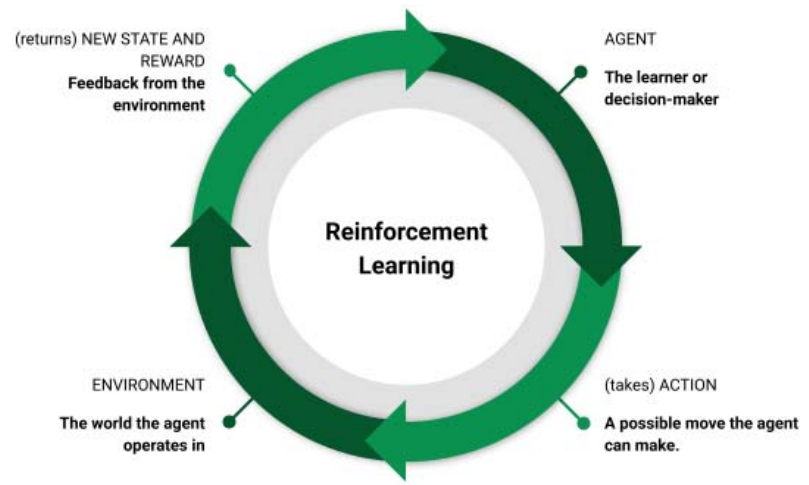
Agent: The learner or decision-maker (e.g., a robot).

Environment: The world the agent operates in (e.g., a maze).

Action: A possible move the agent can make.

Reward: Feedback from the environment (positive or negative).

Policy: The agent's "strategy" for choosing actions to maximise its cumulative reward.



Foundational Concepts of RL

- **Agent:** The learner or decision-maker. It's the algorithm you are training. (e.g., the dog, the AI playing chess).
- **Environment:** The world the agent operates in and interacts with. (e.g., the room, the chessboard).
- **State:** A snapshot of the environment at a particular moment. (e.g., the dog's position in the room, the current layout of pieces on the chessboard).
- **Action:** A possible move the agent can make in a given state. (e.g., "walk forward," "move pawn to e4").
- **Reward:** The feedback from the environment after an action. It can be positive (a treat) or negative (a penalty, like losing a piece in chess). The agent's goal is to maximise its **cumulative reward**—the total reward over the long run, not just the next immediate reward.
- **Policy:** The agent's strategy or "brain." It's what the agent uses to decide which action to take from a given state. The goal of training is to find the optimal policy.

The Exploration vs. Exploitation Trade-off

This is a fundamental dilemma that every RL agent faces. To maximise its long-term reward, the agent must balance two competing goals:

- **Exploitation:** Using the knowledge it already has to take actions that it knows will lead to a good reward.
 - *Analogy:* Going to your favorite restaurant. You know the food is good, so it's a safe, reliable choice.
- **Exploration:** Trying new, random actions to see what happens. This might lead to a worse outcome, but it could also lead to the discovery of a much better strategy.
 - *Analogy:* Trying a brand-new restaurant. It might be terrible, but you might also discover a new favourite that's even better than your old one.

A successful RL agent cannot just exploit what it knows; it must also intelligently explore to find better strategies. Finding the right balance is one of the biggest challenges in reinforcement learning.

Flashcards

Use this Quizlet to learn all the key terms in this section.



Ready to play?

Match all the terms with their definitions as fast as you can. Avoid wrong matches, they add extra time!

Start game

[View this flashcard set](#)

Choose a Study Mode 

Knowledge Check: Multiple Choice Quiz

Test your understanding of the core concepts.

1

Reinforcement Learning is a paradigm where an agent learns by:

- A. ☐ Reusing a pre-trained model on a new task.

B. ☐ Studying a large, labelled dataset.

C. ☒ Interacting with an environment through trial and error.

D. ☐ Finding hidden clusters in unlabeled data.

2

In the RL framework, what is the term for the learner or decision-maker?

A. ☒ The Agent

B. ☐ The Policy

C. ☐ The Environment

D. ☐ The Reward

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The primary goal of an RL agent is to learn a policy that maximises its:

A. ☒ Cumulative reward over time.

B. ☐ Immediate, one-time reward.

C. ☐ Exploration rate.

D. ☐ Number of possible actions.

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A robot is learning to navigate a maze. The layout of the maze at a specific moment is an example of a:

A. ☒ State

- B. ☐ Reward
- C. ☐ Policy
- D. ☐ Action

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The "exploration vs. exploitation trade-off" refers to the dilemma between:

- A. ☐ Using a supervised model versus an unsupervised one.
- B. ☒ Using a known good strategy versus trying a new, unknown strategy.
- C. ☐ Storing data in the cloud versus on an edge device.
- D. ☐ Choosing a fast algorithm versus an accurate one.

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Which of the following is the best example of a Reinforcement Learning problem?

- A. ☒ Training an AI to master the game of Go by playing against itself.
- B. ☐ Classifying customer reviews as positive or negative.
- C. ☐ Predicting the price of a house based on its size and location.
- D. ☐ Grouping shoppers into different segments based on purchase history.

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In RL, the feedback signal from the environment that tells the agent how good its last action was is called the:

- A. ☐ State

B. ☒ Reward

C. ☐ Policy

D. ☐ Antecedent

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An agent's strategy for choosing which action to take from a given state is known as its:

A. ☐ Environment

B. ☐ Reward

C. ☒ Policy

D. ☐ State

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When an RL agent chooses to try a new, random move in a game instead of the move it thinks is best, it is performing:

A. ☐ Classification

B. ☐ Inference

C. ☐ Exploitation

D. ☒ Exploration

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The agent-environment interaction is best described as:

A. ☐ A one-time data transfer.

B. ☐ A continuous feedback loop.

C. ☐ A linear process.

D. ☐ A supervised training method.

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